

Diego Martínez (cv)

Institution *KU Leuven*
Position *Postdoctoral researcher*
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WORK EXPERIENCE

- FWO** FWO senior postdoctoral researcher **University:** *KU Leuven, Department of Mathematics*
Supervisor: Prof. Gábor Szabó from October 2025 until now
- Postdoc** Postdoctoral researcher **University:** *KU Leuven, Department of Mathematics*
Supervisor: Prof. Gábor Szabó from September 2024 until September 2025
- Postdoc** Postdoctoral researcher **University:** *University of Münster, Mathematical Institute*
Supervisor: Prof. Wilhelm Winter from October 2021 until August 2024
- PhD** Ph.D. student **Institutions:** *Universidad Carlos III de Madrid and Instituto de Ciencias Matemáticas*
Supervisor: Prof. Fernando Lledó from April 2017 until March 2021
- Master** Master's student **University:** *Universidad Autónoma de Madrid, Departamento de Matemáticas*
Supervisor: Prof. Fernando Lledó from September 2016 to August 2017

EDUCATION

- Ph.D.** University: *Universidad Carlos III de Madrid (UC3M) and Instituto de Ciencias Matemáticas (ICMAT)*
Supervisor: *Prof. Fernando Lledó*, Program: *Ph.D. in Mathematics* (international distinction)
Thesis: *Amenability and coarse geometry of (inverse) semigroups and C^* -algebras*
Grade: *excellent - cum laude* from September 2016 to March 2021
- Master** University: *Universidad Autónoma de Madrid (UAM)*
Supervisor: *Fernando Lledó*, Program: *Master Program in Mathematics and Applications* (60 ects)
Thesis: *Amenability, paradoxical decompositions and applications*
Grade: *10/10* from September 2015 to July 2016
- Degree** University: *Universidad Autónoma de Madrid (UAM)* from September 2010 to June 2015
Program: *Graduate program in Computer Engineering*
- Degree** University: *Universidad Autónoma de Madrid (UAM)* from September 2010 to June 2015
Program: *Graduate program in Mathematics*
Thesis: *Entropía y códigos de Huffman* (in Spanish, **grade 10/10**)

PUBLICATIONS

- [15] D. Martínez, *A Baum-Connes assembly map for essential semigroup crossed products*, <https://arxiv.org/abs/2606.24502> (2026).
- [14] D. Martínez, *On the relation between the product of KK-groups and the KK-group of the product*, <https://arxiv.org/abs/2606.20181> (2026).
- [13] Becky Armstrong, Lisa Orloff Clark, Astrid an Huef, D. Martínez and I. Tolich, *A Dichotomy for Inverse-Semigroup Crossed Products via Dynamical Cuntz Semigroups*, accepted for publication in Münster J. Math. (2026).

- [12] D. Martínez and N. Szakács, *Algebraic singular functions are not always dense in the ideal of C^* -singular functions*, Bull. London Math. Soc. (2025) DOI: 10.1112/blms.70381.
- [11] D. Martínez and F. Vigolo, *A rigidity framework for Roe-like algebras*, <https://arxiv.org/abs/2403.13624> (2024).
- [10] A. Buss and D. Martínez, *Essential groupoid amenability and nuclearity of groupoid C^* -algebras*, J. Funct. Anal. **290** (2026) 111427 (pp. 54) DOI: 10.1016/j.jfa.2026.111427.
- [9] D. Martínez and F. Vigolo, *Roe algebras of coarse spaces via coarse geometric modules*, JNCG (2025) DOI: 10.4171/JNCG/642.
- [8] D. Martínez and F. Vigolo, *C^* -rigidity of bounded geometry metric spaces*, Publ. Math. IHES (2025) DOI: 10.1007/s10240-025-00155-3.
- [7] Y. C. Chung, D. Martínez and N. Szakács, *Quasi-countable inverse semigroups as metric spaces, and the uniform Roe algebras of locally finite inverse semigroups*, to appear in Groups, Geom. Dyn. (2025).
- [6] A. Buss and D. Martínez, *Approximation properties of Fell bundles over inverse semigroups and non-Hausdorff groupoids*, Adv. Math. **431** (2023) 109251 (pp. 54) DOI: 10.1016/j.aim.2023.109251.
- [5] D. Martínez, *A note on the quasi-diagonality of inverse semigroup reduced C^* -algebras*, J. Oper. Th. **90** (2023) (pp. 17) DOI: 10.7900/jot.2021nov29.2424.
- [4] F. Lledó and D. Martínez, *A note on commutation relations and finite dimensional approximations*, Expo. Math. (2022) DOI: 10.1016/j.exmath.2022.08.004.
- [3] F. Lledó and D. Martínez, *The uniform Roe algebra of an inverse semigroup*, J. Math. Anal. Appl. **499** (2021) 124996 (pp. 28) DOI: 10.1016/j.jmaa.2021.124996.
- [2] P. Ara, F. Lledó and D. Martínez, *Amenability and paradoxicality in semigroups and C^* -algebras*, J. Funct. Anal. **279** (2020) 108530 (pp. 43) DOI: 10.1016/j.jfa.2020.108530.
- [1] F. Lledó and D. Martínez, *Notions of infinity in quantum physics*, 60 Years Alberto Ibort Fest Classical and Quantum Physics, Chapter 14, Springer (2019) (pp. 14) DOI: 10.1007/978-3-030-24748-5_14.

SELECTED INTERNATIONAL TALKS

- *On induced local symmetries of the Cuntz semigroup arising from actions*,
Workshop on Cuntz semigroups and dynamics, Workshop, Gothenburg, Sweden Sept. 2026
- *A Gelfand-type duality and C^* -rigidity for metric spaces*,
Groups, dynamics and operator algebras, Conference, Glasgow, United Kingdom Aug. 2026
- *Rigidity and outer automorphisms of Roe algebras*,
Operator Algebras and Applications, Mini-Workshop, Erlangen, Germany July 2025
- *Finite distortion operators, Roe algebras, and coarse invariants*,
Functional and Metric Analysis, and their Interactions, BIRS, Spain May 2025
- *Cartan subalgebras, cp maps, and their inductive limits*,
Cartan Subalgebras in Operator Algebras, and Topological Full Groups, BIRS, Canada Nov. 2024
- *Amenability of essential inverse semigroup C^* -algebras*,
Plenary talk, SEMigroups, grOUpoids and C^* -algebras, KIAS, Korea July 2024
- *Roe algebras are rigid*,
Analysis Seminar, Newcastle University, UK May 2024
- *Amenability notions and Fell bundles*,
Combinatorial $*$ -algebras, MFO, Oberwolfach March 2024
- *Large-scale geometry of group-like objects*,
Colloquium, University of Göttingen May 2023
- *Minicourse on C^* -algebras, their diagonals, and (inverse) semigroup actions*,
Topics in Geometric Semigroup Theory, University of York June 2022

- *Exactness and geometric properties of inverse semigroups*,
Non-commutativity in the North, Göteborgs Universitet March 2022
- *C^* vs coarse properties of inverse semigroups*,
Analysis online seminar, University of Glasgow Oct. 2021
- *Schützenberger graphs and property A, or how to see exactness of the reduced semigroup C^* -algebra*,
 C^* -algebras and geometry of groups and semigroups, University of Oslo March 2021
- *Non-group like metrics appearing in group-like objects*,
UK Virtual Operator Algebras Seminar, United Kingdom Dec. 2020
- *Some quasi-isometric invariants for inverse semigroups*,
York semigroup (online talk), University of York, United Kingdom May 2020
- *Coarse geometry and inverse semigroups*,
Colloquium talk, University of Montana, USA Nov. 2019
- *Traces in inverse semigroup Roe algebras*,
YMC*A, University of Copenhagen, Denmark Aug. 2019
- *Amenability in inverse semigroups*,
SandGAL-2019, University of Cremona, Italy June 2019
- *Amenability in inverse semigroups and C^* -algebras*,
II Joint Meeting Spain-Brazil in Mathematics, Universidad de Cádiz, Spain Dec. 2018

TEACHING

- **Functional Analysis**, at KU Leuven, Fall 26
- **Master's Thesis**, at KU Leuven, Fall 25/Spring 26
- **Master Thesis Reader**, at KU Leuven, Spring 2025
- **Bachelor's Final Degree Project**, at KU Leuven, Spring 2026
- **Functional Analysis**, at KU Leuven, Fall 25
- **Bachelor's Final Degree Project**, at KU Leuven, Spring 2025
- **Master Thesis Reader**, at KU Leuven, Spring 2025
- **Student Teaching Assistant**, at KU Leuven, Fall 2024
Differentiaalvergelijkingen (differential equations), Reeks 11+12. Bachelor in Bio-engineering.
- **Cartan Subalgebras**, at the University of Münster Fall 2022
Reading seminar
- **TFG committee member**, at UC3M Fall 2020
- **Quasi-diagonality and the Tikuisis–White–Winter Theorem**, at Texas A&M Fall 2019
Reading seminar
- **TFG committee member**, at UC3M Spring 2019
- **Student Teaching Assistant**, at UC3M Fall 2018
Cálculo I (calculus I), group 45. Bachelor in Aerospace Engineering
- **Student Teaching Assistant**, at UC3M Fall 2017
Álgebra lineal I (linear Algebra I), group 49. Bachelor in Energy Engineering

PARTICIPATION IN COMMITTEES

- Member of the **Party Committee**,
at KU Leuven 2024 – present.
- Member of the **Early Career Committee**,
at the University of Münster 2023 – 2024.

REVIEWER OF

- zbMATH Open 2022 – present.
- MathSciNet 2023 – present.

SCHOLARSHIPS AND AWARDS

- **FWO Senior Postdoctoral Researcher** (FWO) - Research Foundation, Flanders 2025 – now
Grant for senior postdoctoral researchers in the Flanders region, Belgium. Allows to conduct research as one sees fit. Funds all of the researcher's salary and travel funds.
- **Formación de personal investigador** (FPI) - Severo Ochoa 2017 – 2021
Grant supporting the fulfillment of the PhD from the MINECO (Spanish Ministry of Economy)
- **Mathematics Department Scholarship** (UAM) 2015 – 2016
Grant intended to support the study of the Master in Mathematics (one of the top 5)
- **BIT Summer School in Beijing** 2012
Grant awarded to pay for an intensive course on Chinese culture and language

RESEARCH VISITS

- **University of Wollongong**, Australia Oct.–Nov. 2023
- **Texas A&M**, USA Aug.–Dec. 2019
Department of Mathematics
- **UAB**, Spain 2017 & 18
Centre de Recerca Matemàtica
- **University of Münster**, Germany May 2016
Institute of Theoretical Mathematics

ORGANIZER OF

- **Junior Seminar**, ICMAT Sep. 2018–June 2021
Local seminar for Ph.D. students
- **OAGAQI** thematic semester, March–June 2019
Thematic semester about operator algebras, groups and applications to quantum information
- **JAE School**, ICMAT June 2017
School that serves as an introduction to research for undergraduate students
- **Simon Marais Math Competition**, KU Leuven Fall 2026

OUTREACH

- **Outreach newspaper** article in *El País* 2019
– *Ganar juegos usando la mecánica cuántica*
- **Outreach newspaper** article in *El País* 2017
– *Dándole vueltas al infinito*

LANGUAGES

- **Spanish:** Native
- **English:** Proficient (C2)
- **German:** Beginner (A2)
- **Dutch:** Beginner (A1)